Price Prediction of Ford Cars Applying Multiple Machine Learning Methods

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Abstract: The growing supply of used cars and lower prices compared with their counterparts make the used car market highly competitive. Such a dynamic and sophisticated market underscores the necessity for accurate price prediction, which is crucial for both buyers and sellers. Nowadays, the popularisation of online databases and advanced machine learning techniques have made price prediction based on machine learning models an obvious trend in the used car trading industry. This study delves into the performance evaluation of multiple machine learning methods for predicting used Ford car prices. Utilizing a comprehensive dataset from Kaggle, encompassing 17,966 entries with nine distinct features, the researcher employed a battery of regression models, including linear regression (LR), decision tree (DT), random forest (RF) and eXtreme Gradient Boosting (XGBoost). The approach involved rigorous feature engineering, model training and cross-validation evaluation, employing Mean Squared Error (MSE), Mean Absolute Error (MAE), and R-squared (R^2) as key performance indicators. The results indicate that XGBoost and RF surpass traditional models in predictive accuracy, with XGBoost demonstrating the highest R^2 value of 0.9347. This study compares the performance of several widespread models and offers practical implications for stakeholders seeking to enhance transactional outcomes through data-driven pricing strategies.

1 INTRODUCTION

Ford Motors is the second largest automobile manufacturer in the United States. Since 1903, the company has launched countless iconic models like Mustang, Bronco, Cortina and Transit. As the first motor manufacturer that implements assembly lines and vertical integration into industrial production, Ford contributes a lot to the modern car industry (Mead & Brinkley, 2003). Trusted quality and moderate depreciation rate make used Ford cars popular in the market.

Over the past few decades, the flourishing second-hand car market has become a vibrant sector within the automotive industry, offering consumers a reasonably-priced alternative to a brand-new counterpart. According to Edmunds, an online resource that provides car buying information and vehicle sales data, approximately 39 million used vehicles were sold in the U.S. throughout the year 2021. Manheim, a global provider of vehicle remarketing services, introduced Manheim Used Vehicle Value Index to reflect the fluctuation of America's used car prices (Manheim, 2023). The index showed an upward trend in price over the last five years.

Accurate price prediction for used cars is not only crucial for buyers seeking value but also essential for sellers aiming to maximize their profits. As the volume of transactions continues to grow, the precision of used car pricing has garnered considerable interest, prompting researchers to pursue better performance of various predictive models.

The complexity of the used car market requires advanced and sophisticated analytical tools to evaluate the fair value of vehicles based on multi-dimensional factors. The manufacturer who produces the vehicle, mileage, transmission and fuel economy all contribute to the training process of machine learning models. The application of data-driven models has transformed the way prices are estimated, moving from heuristic methods to evidence-based approaches. Cui et al. (2022) discussed the integration

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of machine learning algorithms in predicting used car prices, which is vital in enhancing the accuracy and efficiency of predictions.

Previous scholarly endeavours in the domain of used car price prediction have primarily leveraged models such as linear regression (LR), decision tree (DT), and neural networks. LR models, despite their simplicity, offer a baseline for understanding the relationship between car attributes and their actual price (Dahal, 2023). Jin (2021) constructed DT models, including ensemble methods such as random forest (RF) in their research. Ensemble models were also widely applied due to their ability to deal with feature interactions under non-linear correlations and provided robust predictions. Neural networks, particularly deep learning architectures, have demonstrated remarkable potential in capturing complex patterns, especially when applied to datasets on a large scale (Huang et al., 2023).

The evolution of these models reflects the growing need for more nuanced and dynamic pricing strategies. However, the heterogeneity of used cars and the rapid pace of technological advancements in the automotive industry present ongoing challenges. The quest for more accurate and efficient predictive models continues, with recent studies exploring ensemble methods and hybrid approaches that combine the strengths of multiple algorithms.

The previous study has shown the necessity of combining multiple regressors to better evaluate used cars' prices. This paper aims to evaluate the performance of different regressors and discover critical factors that influence used cars' value. The author will first introduce the dataset used during the study, and then discuss the main methodologies and techniques applied. At the end, the author will offer an overview of the research and discuss limitations and possible optimization methods.

2 DATASETS AND METHODS

2.1 Data Collection and Description

The dataset used in the research is obtained through Kaggle. It highlights 9 features that might influence the pricing procedure of used Ford vehicles. Those features include the specific model of the car and its production year, both of which depict the overall price section of certain models. However, prices of the same models vary significantly as each car has different mileage, engine size, transmission, fuel consumption and tax. The dataset collects detailed information on 17,966 used Ford vehicles priced in U.S. dollars. The introduction of characteristic properties used during the research is shown in Table 1.

Table 1: Basic characteristic properties of the data.

ID	Features	Type
1	model	object
2	year	int64
3	price	int64
4	transmission	object
5	mileage	int64
6	feulType	object
7	tax	int64
8	mpg	float64
9	engineSize	float64

Among the features mentioned in Table 1, transmission tells the researcher what kind of gearbox the car is equipped with, including manual, semi-auto and automatic. Mpg stands for mile per gallon, which is an indicator of vehicle's fuel economy. The figure of engine size indicates how many litres the engine has in swept volume.

To better carry out the following research, the author first conducted the Exploratory Data Analysis (EDA). The data chosen doesn't have null values and the size of duplicate records is relatively small. However, it is worth noting that the distribution of samples on model, transmission and fuel type is highly imbalanced. To improve the performance of models applied, the paper dealt with those rare data through feature engineering.

2.2 Feature Engineering

Before model training, it is vital to ensure data quality through data pre-processing. To optimize the performance of selected models, the author processes the data with the following steps. The first step is to divide characteristic properties into two groups, numerical and categorical. The classification standard is slightly different from the ordinary one, that is the researcher treats engine size and tax as categorical features as they cluster at a few discrete values.

After grouping, the researcher focuses on rare data among the categorical features. To reduce the disturbance of those records, a threshold of 0.05 is set in the paper. Features with a ratio less than 5% will be added to a new group called "Other". Figure 1 shows the distribution of values on 4 features, including model, engine size, fuel type and tax. All kinds of transmission options meet the threshold of 0.05 so it is not necessary to update the bar chart.

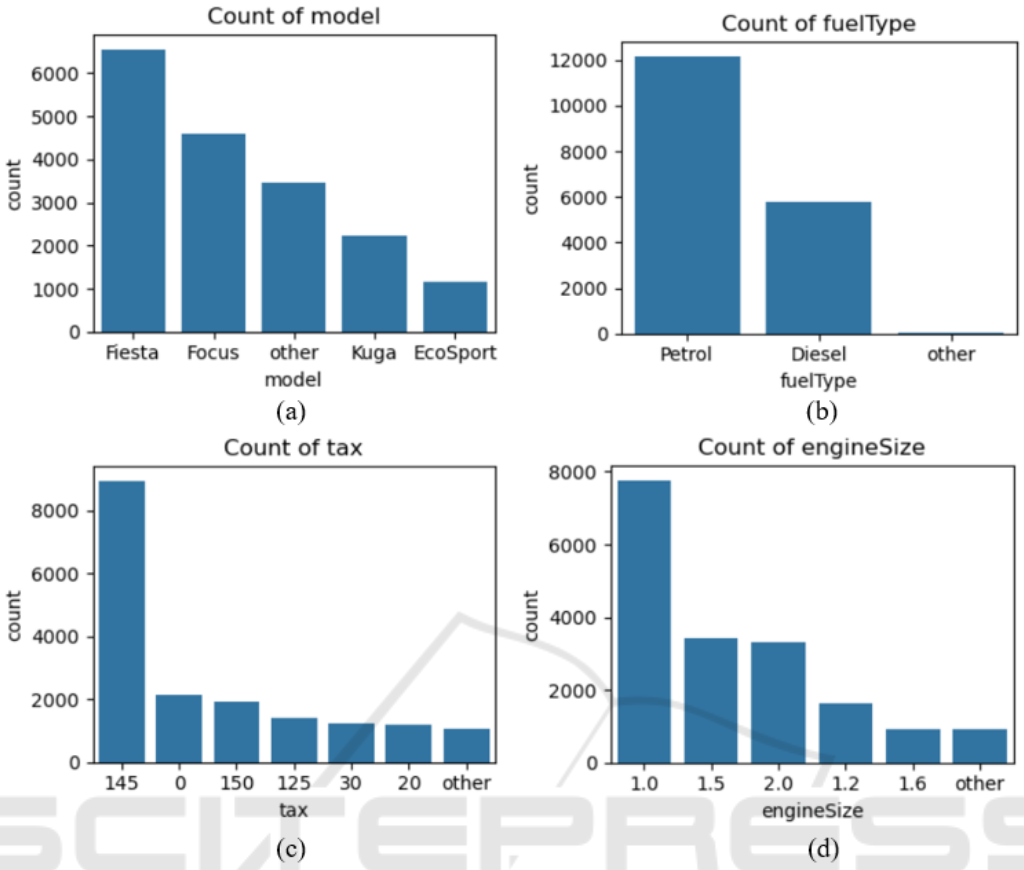


Figure 1: Distribution of values across selected features. (a) Distribution of Model types, (b) Distribution of Fuel types, (c) Distribution of Tax, (d) Distribution of Engine Size (Photo/Picture credit: Original).

For each categorical feature f, this paper applies functions to perform one-hot encoding. This function converts the categorical feature into a series of binary columns, where each column represents a category, and each sample has only 1 in the corresponding column, with 0s in all others. After the encoding procedure, the author checked each option's correlation with the price tag of the vehicle. By sorting them into descending order, the study dropped three low-correlated features with an influence coefficient below 0.05. As for the numerical features, this study has found that none of the three features follows normal distribution according to respective count figures. In this case, it is essential to apply scalers to reduce the size of evaluation metrics in the following tests. This paper applies MinMaxScaler to process corresponding values.

2.3 Linear Regression

Linear regression (LR) is a statistical methodology employed to model the correlation between the variable to be predicted and a set of independent variables. It searches for the best-fitting line through data points under the supposition that the variables have linear correlations. The coefficient of determination is used to gauge the strength of the model, which is usually expressed as an equation. LR is extensively employed in various domains for forecasting, inference, and prediction.

2.4 Decision Tree & Random Forest

Decision Tree (DT) is a machine-learning algorithm based on tree-like model to predict continuous outcomes. It works by splitting the data into subsets based on feature values, creating a tree of decisions that ultimately yield a prediction. This method is robust, easy to interpret, and capable of handling non-linear relationships and feature interactions.

Random Forest (RF) is a versatile ensemble learning method that effectively addresses regression challenges by combining the predictions of multiple decision trees (Fawagreh et al., 2014). It harnesses the power of bagging (Bootstrap Aggregating) to enhance the stability and accuracy of the model.

The training procedure of each tree involves different bootstrap samples of the data and the model makes its final prediction by averaging the individual tree predictions, thereby reducing the variance and preventing overfitting.

The algorithm incorporates randomness by selecting a random subset of features to consider at each split in the trees, which promotes diversity among the trees and improves the model's generalization capabilities. RF is particularly useful for high-dimensional datasets and is known for its robustness to outliers and noise.

In the research, the author applied early stopping to select a suitable value for n_estimators, which indicates the number of DT. Normally speaking, implementing more DT enables the model to achieve better performance. However, after the n_estimator reaches the performance saturation point, adding more trees to the ensemble no longer leads to significant improvements in the model's predictive performance. Early stopping assists the computer in making the trade-off between computation time and model's performance by stopping the optimization process when the improvement is lower than the significance threshold set before the evaluation.

2.5 Extreme Gradient Boosting

The eXtreme Gradient Boosting (XGBoost) is another efficient ensemble method designed for regression and classification tasks (Chen & Guestrin, 2016). It builds models in an additive manner by focusing on minimizing a loss function through a gradient-boosting framework. XGBoost employs a tree-based approach, enhancing the predictive accuracy by scaling the weak learners and adding them to the final prediction. It also introduces regularization to control overfitting, making it robust for a wide range of data sets (Yang et al., 2023). With its user-friendly interface and high performance, XGBoost has become a go-to tool for data scientists and machine learning practitioners in various competitions and industries.

3 RESULTS AND DISCUSSION

3.1 Evaluation Metrics

The performance of selected models in this study was assessed using Mean Squared Error (MSE), Mean Absolute Error (MAE), and R-Square (R^2).

MSE is the average squared difference between the estimated values and the actual values. The formula for MSE is shown in Equation (1).

$$MSE = \frac{1}{n}\sum_{i=1}^{n}\left(Y_i - \hat{Y}_i\right)^2 \quad (1)$$

In Equation (1), $\hat{Y}_i$ stands for the actual value while Y_i means predicted value. The character n is the size of observations involved.

MAE measures the average magnitude of errors among predictions without considering their direction. Therefore, the figure is always non-negative. The formula for MAE is shown in Equation (2).

$$MAE = \frac{1}{n}\sum_{i=1}^{n}\left|Y_i - \hat{Y}_i\right| \quad (2)$$

R^2 is a metric that indicates how effectively the model is likely to predict future outcomes. It demonstrates the proportion of the variance in the dependent variable which can be predicted from the independent variables. According to Chicco et al. (2021), R^2 is capable of overcoming the interpretability limitations of MSE and MAE. The formula for R^2 is shown in Equation (3).

$$R^2 = 1 - \frac{\sum_{i=1}^{n}\left(Y_i - \hat{Y}_i\right)^2}{\sum_{i=1}^{n}(Y_i - \bar{Y}_i)^2} \quad (3)$$

3.2 Model Training

Before the author fits the data into the DT, LR is applied to set a baseline, achieving an R^2 of 0.7779. The result of DT significantly improves the performance with the MSE in 8e-3, MAE in 0.0201 and R^2 in 0.8832.

To further optimize the performance of the model, this study applies the method of Bootstrap Aggregating by increasing the number of DT during the predicting procedure. The approach has proved effective in strengthening the model's robustness and preventing overfitting. To find a suitable number of trees acquired, the researcher uses an iterative algorithm that compares the cross-validation score between two nearby rounds. The improvement threshold is set at 0.1%, indicating that the algorithm will stop after the model fails to reach the improvement threshold for 5 consecutive tries. The cross-validation score is calculated based on model's R^2 figure. Eventually, the performance saturated after 22 trees. The R^2 figure reached 0.9222. Table 2 shows the performance evaluation of three regression trained so far.

Table 2: Performance Evaluation of three regression.

ID	Model	MSE	MAE	R^2
1	Linear Regression	1.68e-3	0.0283	0.7779
2	Decision Tree	8.81e-4	0.0201	0.8832
3	Random Forest	5.86e-4	0.0166	0.9222

Table 3: Performance Evaluation of four regression.

ID	Model	MSE	MAE	R^2	CV Score
1	LR	1.68e-3	0.0283	0.7779	79.1941
2	DT	8.81e-4	0.0201	0.8832	87.9012
3	RF	5.86e-4	0.0166	0.9222	91.7158
4	XGBoost	4.93e-4	0.0152	0.9347	93.0520

Then researcher generates importance values of different features by implementing functions of DT. Among all the features, the top seven characteristic properties show a higher correlation with the price of used vehicles. Figure 2 demonstrates their important value to the price of cars.

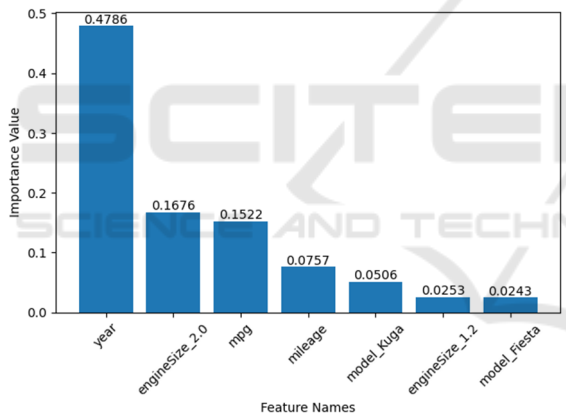


Figure 2: Top 7 features that influence vehicle pricing (Photo/Picture credit: Original).

From Figure 2, it is obvious that consumers consider production year as the most vital factor when purchasing used vehicles, followed by engine size of 2.0 and mpg. The study also finds that consumers may prefer an engine size of 2.0 to 1.2 or other options. Ford Kuga generally has a higher price according to both Figure 2 and the original dataset.

The study then trains the XGBoost model and makes it continue to optimize its performance based on the results of previous models. Yet the improvement of the model will reach a bottleneck where further optimization is less evident but time-consuming. To avoid the situation, an early-stopping algorithm is applied with the threshold set at 5 rounds.

The final iteration result indicates that the model scores 0.9347 in R^2. Figure 3 shows the change of Root Mean Squared Error (RMSE) figure as iteration grows.

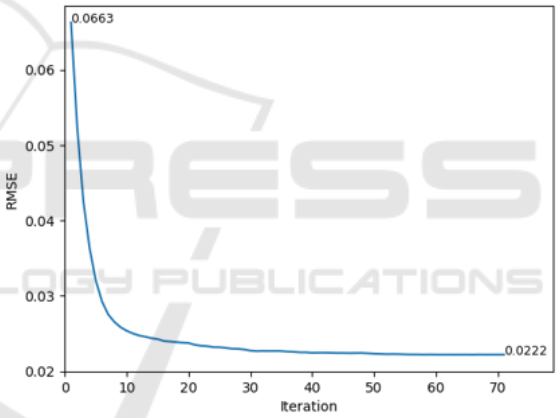


Figure 3: RMSE value as the iteration grows (Photo/Picture credit: Original).

Table 3 compares the performance of four selected models. CV Score demonstrates their performance on cross-validation accuracy judged by the R^2 figure.

From the results shown above, it is obvious that XGBoost outperforms its counterpart at the price prediction of the given dataset. Closely followed by RF. Despite its simplicity, LR still achieves a moderate performance.

4 CONCLUSIONS

Tens of thousands of used vehicles are traded every day around the world, making timely and accurate

price predictions essential to boost the efficiency and profit of the industry. Therefore, applying machine learning methods to assist participants with decision-making is promising. This study offers insightful information about used automobile pricing predictions. Not only has it found the key features which might affect the price of cars, but also shown the preference of customers for those features. For example, the importance values of various features indicate that a 2.0-litre engine affects the price more than its 1.2-litre counterparts. This study also demonstrates the effectiveness of used car price prediction based on machine learning methodology, with XGBoost and RF showing superior performance. The cross-validation scores indicate that these models offer a high level of accuracy and are capable of providing reliable price predictions for used cars. However, the research has some limitations at the present stage. Firstly, the car market is a sophisticated system and the detailed information on models involves complicated terms. The dataset selected focused on a single manufacturer Ford and simplified the issue. More work on data pre-processing is acquired to predict data of a much larger scale, such as annual domestic used car trading. Apart from that, more features involved will inevitably lead to the Multi-collinearity problem. In that case additional machine learning methods need to be implemented to handle that issue.

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